

Semantics about soil organic carbon storage: DATA4C+, a comprehensive thesaurus and classification of management practices in agriculture and forestry

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ABSTRACT ORIGINAL

Identifying the drivers of soil organic carbon (SOC) stock changes is of the utmost importance to contribute to global challenges like climate change, land degradation, biodiversity loss, or food security. Evaluating the impacts of land use and management practices in agriculture and forestry on SOC is still challenging. Merging datasets or making databases interoperable is a promising way, but still has several semantic challenges. So far, a comprehensive thesaurus and classification of management practices in agriculture and forestry has been lacking, especially while focusing on SOC storage. Therefore, the aim of this paper is to present a first comprehensive thesaurus for management practices driving SOC storage (DATA4C+). The DATA4C+ thesaurus contains 224 classified and defined terms related to land management practices in agriculture and forestry. It is organized as a hierarchical tree reflecting the drivers of SOC storage. It is oriented to be used by scientists in agronomy, forestry, and soil sciences with the aim of uniformizing the description of practices influencing SOC in their original research. It is accessible in Agroportal (<http://agroportal.lirmm.fr/ontologies/DATA4CPLUS>, last access: 24 March 2022) to enhance its findability, accessibility, interoperability, and reuse by scientists and others such as laboratories or land managers. Future uses of the DATA4C+ thesaurus will be crucial to improve and enrich it, but also to raise the quality of meta-analyses on SOC, and ultimately help policymakers to identify efficient agricultural and forest management practices to enhance SOC storage. © Copyright: